

nRF54L Series Hardware Design

Guideline

Contents

	Revision history	iii
1	Introduction	4
2	High-frequency crystal oscillator	5
	2.1 HFXO configuration with internal capacitors.	5
	2.2 HFXO configuration with external capacitors.	6
	2.3 Select an HFXO crystal.	6
	2.3.1 Crystal performance parameters.	7
	2.4 Startup time for selected HFXO crystal.	9
	2.5 Example case for HFXO crystal selection.	9
	2.6 HFXO software configuration.	10
	2.6.1 HFXO devicetree.	11
	2.6.2 Troubleshooting.	13
	2.7 Test the HFXO crystal.	13
3	Low-frequency crystal oscillator	14
	3.1 LFXO configuration with internal capacitors.	14
	3.2 LFXO configuration with external capacitors.	15
	3.3 Select an LFXO crystal.	15
	3.4 Startup time for selected LFXO crystal.	16
	3.5 Example case for LFXO crystal selection.	17
	3.6 LFXO software configuration.	18
	3.6.1 LFXO devicetree.	18
	3.7 Typical LFXO crystal specifications.	19
4	PCB layout design	20
	4.1 Component placement.	20
	4.2 PCB stack-up.	20
	4.3 DECRF routing.	21
	4.4 Shunt capacitor for QFN.	22
	4.5 HFXO crystal placement and routing.	23
	4.6 LFXO crystal placement and routing.	23
	Glossary	24
	Recommended reading.	25
	Legal notices.	26

Revision history

Date	Description
May 2026	First release

1 Introduction

This document provides guidelines for the hardware design and integration of the nRF54L Series devices. It gives guidelines for the selection of high-frequency and low-frequency crystal oscillators and PCB layout design. The guideline is intended for system integrators and hardware engineers.

2 High-frequency crystal oscillator

All HFXOs in the nRF54L Series use a 32 MHz crystal in fundamental mode.

To use the HFXO, place a 32 MHz crystal between pins **XC1** and **XC2**. The crystal datasheet specifies the load capacitance (C_L), which is assumed to be a parallel capacitance. This means that the load capacitance is the effective capacitance seen by the crystal, which includes everything.

The following figure shows how the 32 MHz crystal is connected to the HFXO.

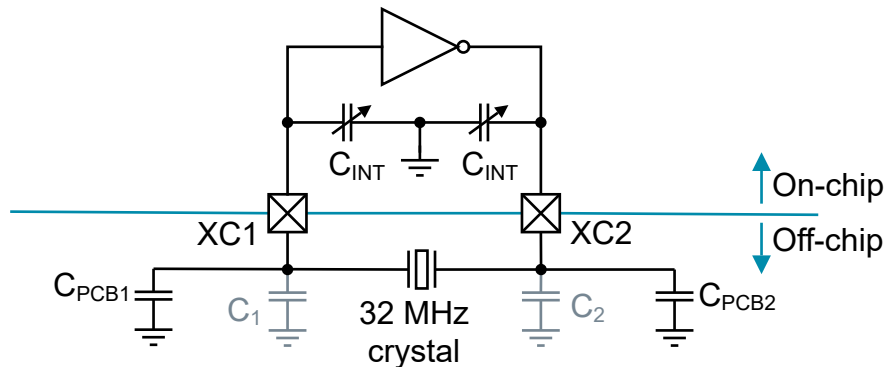


Figure 1: HFXO

The HFXO can be configured to use either the external capacitors C_1 and C_2 or the internal capacitors C_{INT} .

For more information, see [High-frequency \(32 MHz\) crystal oscillator \(HFXO\)](#) in the Datasheet.

2.1 HFXO configuration with internal capacitors

nRF54L Series products have internal programmable capacitors at the input and output of the crystal oscillator nodes.

The internal capacitance values can be adjusted to achieve the overall C_L load needed to satisfy the frequency accuracy. To configure, see [High-frequency \(32 MHz\) crystal oscillator \(HFXO\)](#) in the nRF54L15 Datasheet.

The following figure shows the 32 MHz crystal connected to the high-frequency crystal oscillator with internal capacitors.

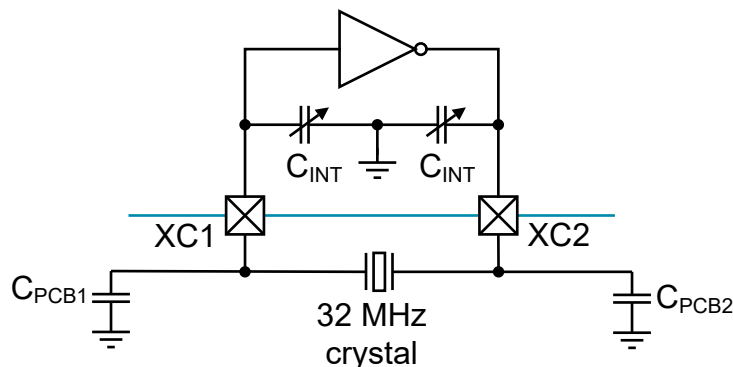


Figure 2: HFXO circuit diagram with internal capacitors only

The load capacitance (C_L) for the internal capacitors is calculated with the following equation.

$$C_L = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C_{PIN} + C_{PCB1}$$

$$C2' = C_{PIN} + C_{PCB2}$$

Figure 3: Load capacitance equation for internal capacitors

C_{PCB1} and C_{PCB2} are stray capacitance on the PCB. C_{PIN} is the pin input capacitance on pins **XC1** and **XC2**, including C_{INT} and other capacitive loadings from the internal pad, on-chip ESD structure, and metal routes.

The internal capacitors must be configured before starting the HFXO. For more information, see the relevant Datasheet for your device.

2.2 HFXO configuration with external capacitors

nRF54L Series products can use external capacitors to provide the required load capacitance (C_L).

For external capacitors, the load capacitance can be calculated with the following equation.

$$C_L = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C_1 + C_{PCB1} + C_{PIN}$$

$$C2' = C_2 + C_{PCB2} + C_{PIN}$$

Figure 4: Load capacitance equation for external capacitors

C_1 and C_2 are the load capacitances of the external capacitors. C_{PCB1} and C_{PCB2} are stray capacitance on the PCB. C_{PIN} is the pin input capacitance on pins **XC1** and **XC2**.

2.3 Select an HFXO crystal

The nRF54L Series *HFXO* supports a range of equivalent series resistance (ESR) and shunt capacitance (C_0) to meet the startup requirements.

The following figure shows a simple model of a crystal. It has R-L-C series components called equivalent series resistance (ESR), motional inductance (L_M), and motional capacitance (C_M). The capacitor in parallel (C_0) is called the shunt capacitance, and it models the package capacitance.

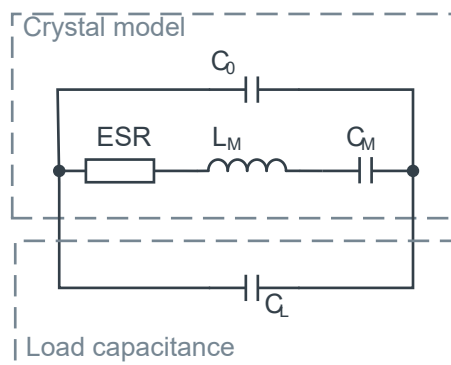


Figure 5: Simplified crystal model

The following figure shows the maximum allowed combinations of ESR and C_0 for different load capacitance (C_L) values. A crystal is supported when its parameters fall on or below the load capacitance line of interest. A crystal is not supported when its parameters are above the line.

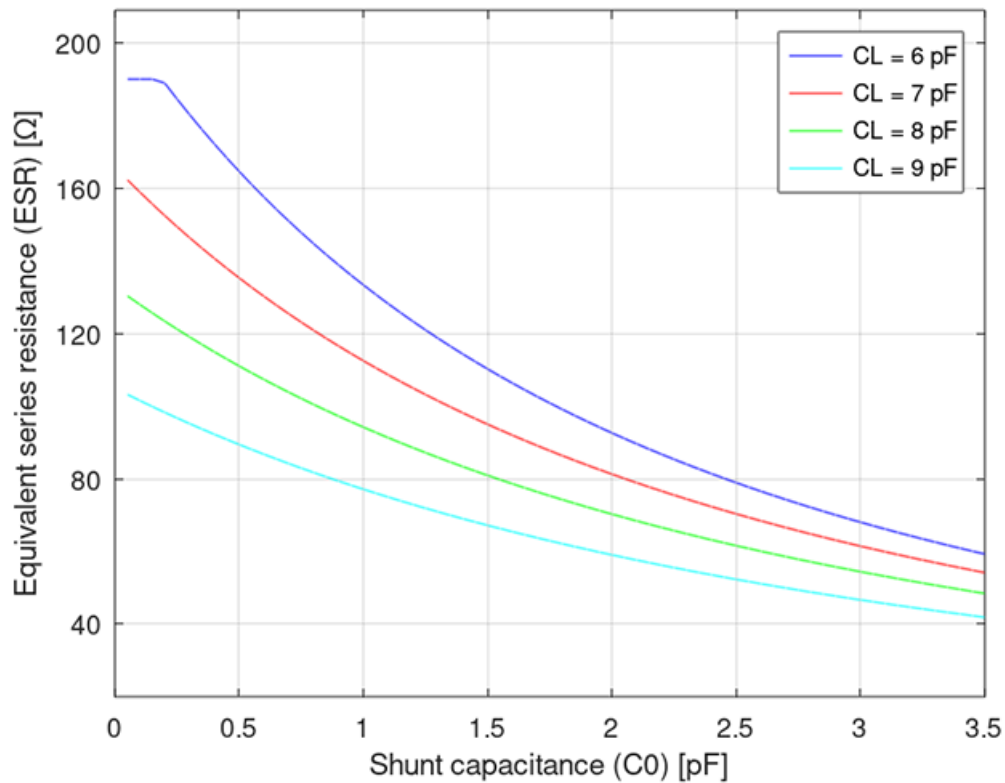


Figure 6: Maximum allowed combinations of ESR and C_0 for different load capacitance (C_L) values

2.3.1 Crystal performance parameters

Information specified in a crystal datasheet typically consists of combinations of the following performance parameters.

Parameter	Symbol	Description
Crystal frequency	f	All nRF54L Series use 32 MHz.
Mode of operation		Fundamental or overtone.
Storage temperature	T_{RST}	Recommended storage temperature range that does not cause damage.
Operating temperature	T_A	Temperature range that covers the given crystal's operation.
Frequency stability		The change in oscillator frequency over time under varying conditions. Consists of the following three major sources: • Manufacture tolerance f_{TOL} (X ppm) • Temperature tolerance (Y ppm) • Aging tolerance (Z ppm) The overall tolerance (ppm) = $X + Y + Z$

Parameter	Symbol	Description
Manufacture tolerance	f_{TOL}	Initial crystal frequency deviation from nominal value due to manufacturing tolerances. Typical range is from ± 10 ppm to ± 50 ppm.
Temperature tolerance		This can be defined as $\pm \text{ppm}/^{\circ}\text{C}$ or $\pm \text{ppm}$ over an operating temperature range.
Aging tolerance	f_{age}	Frequency deviation over time due to aging effects typically specified per year. Generally it is specified for the first year. After that, the rate usually decreases. For example, the tolerance might be ± 2 ppm in the first year and ± 1 ppm or less in the subsequent years. Assumes continuous operation. On/off cycling reduces degradation. The actual value depends on target life and duty cycle.
Motional capacitance	C_M	Capacitance in the series resonant arm. Represents mechanical elasticity.
Motional inductance	L_M	Inductance in the series resonant arm. Represents the effective mass of the vibrating quartz element. For a 32 MHz crystal, typical values range from approximately 2.5 mH to 10 mH.
Equivalent series resistance	ESR	Ohmic resistance of the motional arm in the crystal's equivalent circuit. Affects oscillator startup time and gain margin. A higher ESR results in a lower oscillator gain, longer startup time, lower Q-factor, and degraded phase noise.
Shunt capacitance	C_0	The parallel capacitance between two metal electrodes on the quartz blank. Independent of the mechanical resonance.
Load capacitance	C_L	External capacitance, including stray, required in the oscillator circuit for the crystal to resonate at fundamental frequency. Calculated as parallel capacitance to crystal model.
Q-factor	Q	Indicates quality of the resonance network, E_{STORED}/E_{LOSS}. Can be expressed as follows: $Q = \frac{1}{2\pi f R_1 C_1}$
Drive level	D_L	Maximum power that the crystal can safely dissipate without physical damage. Typically, in the order of 100 μW for a 32 MHz crystal.

Table 1: Crystal performance parameters typically specified in the crystal datasheet

2.4 Startup time for selected HFXO crystal

Estimate the startup time for the nRF54L Series 32 MHz HFXO to ensure correct operation and performance.

Use the following equation, which is based on simulation and modeling results, to estimate the startup time in μs .

$$T_{\text{STARTUP_ESTIMATED}} = L_M \cdot (19.6 \cdot C_0 + 13.5 \cdot C_L - 70) + 256$$

L_M is the crystal's motional inductance, which is specified in mH. C_0 is the shunt capacitance, which is specified in pF. C_L is the parallel load capacitance from the selected crystal's datasheet, which is specified in pF.

2.5 Example case for HFXO crystal selection

The following example checks the parameters for a sample crystal to confirm that the nRF54L Series supports it. The example shows also how to calculate the startup time.

The following table shows the parameters for a sample crystal.

Parameter	Symbol	Specification
Crystal frequency	f_{nom}	32 MHz
Operating temperature	TA	-40°C to 105°C
Maximum drive level	D_L	100 μW (max)
Frequency tolerance	f_{tol}	± 20 ppm
Frequency tolerance over temperature	f_{tem}	± 20 ppm
Frequency aging	f_{age}	± 2 ppm/year (max), 25°C first year
Motional inductance	L_M	6.7 mH
Motional capacitance	C_M	3.7 pF
Motional resistance	ESR, R_1	60 Ω
Load capacitance	C_L	8 pF (normally multiple listed)
Shunt capacitance	C_0	2 pF max

Table 2: Parameters for crystal selection example

For the sample crystal, ESR is 60 Ω , load capacitance (C_L) is 8 pF, and the maximum shunt capacitance (C_0) is 2 pF.

The graph in [Figure 6: Maximum allowed combinations of ESR and \$C_0\$ for different load capacitance \(\$C_L\$ \) values](#) on page 7 shows the maximum allowable shunt capacitance for an ESR of 60 Ω and C_L at 8 pF. In the graph, the maximum shunt capacitance is about 2.7 pF. The maximum shunt capacitance of the sample crystal is 2 pF, so this crystal is suitable for the nRF54L Series.

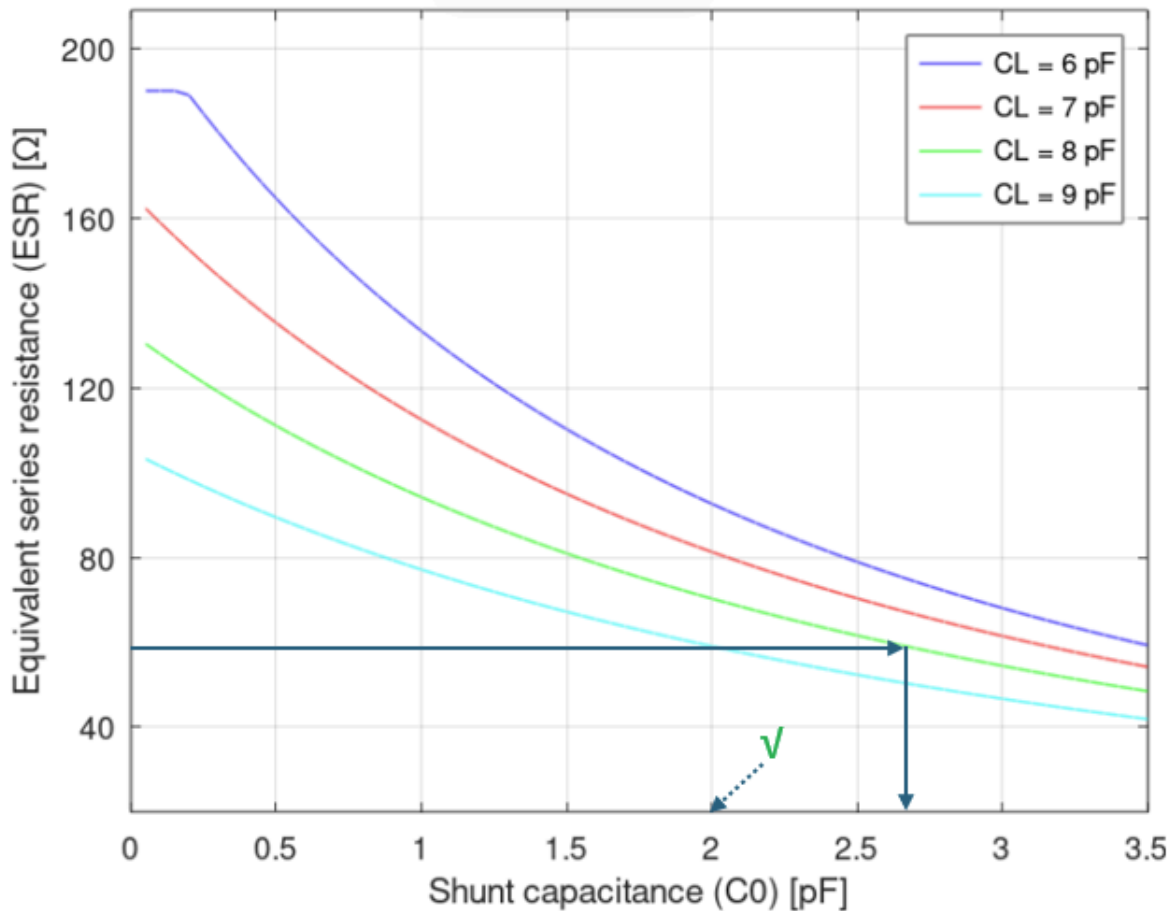


Figure 7: Sample crystal with 60 Ω ESR and 8 pF C_L

An estimate for the startup time of the sample crystal is made with the formula in [Startup time for selected HFXO crystal](#) on page 9. With the parameters of the sample crystal, the startup time can be estimated as follows.

$$T_{\text{STARTUP_ESTIMATED}} = 6.7\text{mH} \cdot (19.6 \cdot 2\text{pF} + 13.5 \cdot 8\text{pF} - 70) + 256$$

This gives an estimated startup time of 0.77 ms for the nRF54L Series HFXO with the sample crystal.

2.6 HFXO software configuration

For the nRF54L Series, software does not speed up a slow crystal. The software ensures correct configuration of the selected hardware and that the firmware waits long enough for the tuned HFXO to be ready. This is important especially for *Bluetooth*® Low Energy and other radio use cases.

After you select the HFXO crystal and load network in the hardware, the software must do the following actions:

- Select internal or external capacitors.
- Program the correct internal capacitor value if the design uses internal capacitors.
- Set a realistic **startup-time-us** budget based on the selected crystal signal.

For a custom board, do not copy an HFXO startup value from a *Development Kit (DK)*. Update the DTS file settings to the actual crystal network on your board.

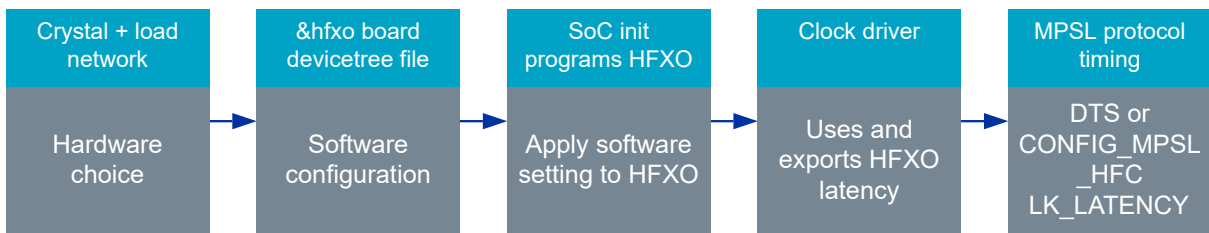


Figure 8: High-level flow from selecting hardware to radio timing inside nRF Connect SDK

2.6.1 HFXO devicetree

For a custom board, configure the *HFXO* settings in the devicetree using the `&hfxo` node. Nordic *DKs* also use this node. In practice, this node is the single source of truth for the board-level HFXO configuration.

The following example sets the load capacitor type, the internal capacitor value, and the startup time for a board HFXO configuration.

```

/* boards/<your_board>.overlay */
&hfxo {
    load-capacitors = "internal";
    load-capacitance-femtofarad = <15000>;
    startup-time-us = <1650>;
};

```

The following table shows the configuration parameters.

Parameter	Function	Comment
load-capacitors	Set the HFXO load capacitor type to <code>internal</code> or <code>external</code> .	Set the load capacitor type to match your hardware design. It should match the actual crystal network on the PCB.
load-capacitance-femtofarad	Use to program the internal capacitor value when internal capacitors are used.	Use only when load-capacitors = "internal" . Use the value chosen for the hardware design.
startup-time-us	Define the HFXO startup budget used by the firmware.	Measure XOSTART-> XOTUNED on real hardware. Ignore the first power-up, then use twice the measured time. The parameter value is specified in μs .
CONFIG_MPSL_HFCLK_LATENCY	Fallback and override path used by the Multiprotocol Service Layer (MPSL) when a DT-based HFXO startup value is not available.	For most boards, keep the DTS startup-time-us as the main board-specific setting.

Table 3: Devicetree configuration parameters

The **load-capacitors** parameter specifies if the board uses the SoC's internal HFXO capacitors or external capacitors. Use **load-capacitance-femtofarad** only when internal capacitors are selected. This value should match the C_{INT} value chosen during the hardware design.

Use the **startup-time-us** parameter as a time budget, not as a tuning control. Do not use the first power-up measurement when selecting the **startup-time-us** value. Set this value based on the measured time in μs from TASKS_XOSTART to EVENTS_XOTUNED. The Datasheet states that XOSTARTED is sufficient for many peripherals. However, some peripherals and functions such as the RADIO and the 32.768 kHz RC calibration should not be enabled before the XOTUNED event. Because of this, **startup-time-us** must be based on the tuned event and not the earlier start event.

If you change the crystal, capacitor mode, capacitor value, layout, voltage range, or temperature range on the board, review the software settings and update when needed.

2.6.1.1 Internal capacitors

When your board uses the internal load capacitors in an nRF54L Series *System on Chip (SoC)*, configure the devicetree to use internal capacitors and set the load-capacitance value to match the value from the hardware design.

Set the devicetree configuration for internal load capacitors according to the following guidelines:

- Use the capacitance value determined during the hardware design.
- Select the closest supported internal capacitor step.
- Verify the configuration through frequency accuracy and *HFXO* startup-time measurements on the final hardware.

The following example sets the load capacitor as internal with a load capacitance of 15000 fF.

```
&hfxo {
    load-capacitors = "internal";
    load-capacitance-femtofarad = <15000>;
};
```

The property **load-capacitance-femtofarad** value specifies the target internal load capacitance used by the *HFXO* driver. During early initialization, the firmware reads this value and configures the internal *HFXO* capacitor bank before the oscillator is started.

This configuration is applied at runtime during boot and is not permanently programmed into the device.

The value used in the software should match the load-capacitance target determined during the hardware design stage. To determine the capacitance value, including crystal load requirements and PCB parasitics, see [HFXO configuration with internal capacitors](#) on page 5.

The **load-capacitance-femtofarad** parameter is specified in femtofarads (fF), while crystal datasheets typically specify capacitance in picofarads (pF). Select a supported value from the node specific properties list in Zephyr™ API reference *nordic,nrf54l-hfxo*.

The **startup-time-us** parameter has no start or end limit set in the firmware. If you set the **startup-time-us** too short, the *HFXO* might start too late. This can cause lost Bluetooth Low Energy events. If you set the **startup-time-us** too long, the *HFXO* runs longer than needed and increases current consumption.

2.6.1.2 External capacitors

When your board uses external load capacitors, configure the devicetree to use external capacitors.

For external load capacitors, the capacitor values are set during the hardware design. The software does not calculate or configure these values. When the software configuration uses external capacitors, the internal capacitor bank inside the SoC is disabled and the value **load-capacitance-femtofarad** is ignored.

For external capacitors, the firmware does not configure the internal *HFXO* capacitor bank. The hardware default is that the internal capacitors are enabled. During startup and after reset, the firmware sets the internal load capacitors to 0. The effective load capacitance seen by the crystal is determined by the

external capacitors (C_{PIN}) parasitic and PCB parasitics. For more information, see [HFXO configuration with external capacitors](#) on page 6.

The following example sets the load capacitor as external.

```
&hfxo {
    load-capacitors = "external";
};
```

2.6.2 Troubleshooting

The following are common issues and oversights that can happen when configuring the software for the HFXO.

Problem	Common cause	Recommendation
Copying the startup-time-us value from a DK to a custom board.	The number might be optimistic or simply wrong for the custom crystal network.	Measure on the final board and replace inherited defaults.
Programming internal capacitance without including PCB stray capacitance.	The crystal might start more slowly or have less frequency margin.	Use the hardware equations and then validate on real hardware. See HFXO configuration with internal capacitors on page 5 and Internal capacitors on page 12.
Trying to fix a hardware mismatch only by increasing startup-time-us .	The design might still be marginal even if software waits longer.	Correct the crystal and load network first. Use startup-time-us as the timing budget, not a hardware fix.

Table 4: Troubleshooting for HFXO software configuration

2.7 Test the HFXO crystal

Use the oscillator output pin **XC2** to test the characteristics of the external crystal oscillator.

Use the following guidelines to test the crystal oscillator's characteristics:

- Use **XC2** which is driven by the on-chip HFXO and less sensitive to added parasitic impedance.
- Choose a probe with minimum loading.
- Use an equivalent electrical model of the probe.
- Design the total load network as close to the original network as possible.
- Do not use these redundancies in the real product due to the impact on performance.

Note: Do not probe on the input of the oscillator amplifier **XC1**. This is a very sensitive, high-impedance node. Any parasitic impedance can change the performance, such as reduce the amplitude, cause a frequency offset, or stop the HFXO oscillation. Any noise injection to this node is amplified with the oscillator amplifier gain.

3 Low-frequency crystal oscillator

For a clock accuracy higher than LFRC, the 32.768 kHz *LFXO* must be used.

To use the *LFXO*, place a 32.768 kHz crystal between pins **XL1** and **XL2**.

The following figure shows how the 32.768 kHz crystal is connected to the *LFXO*.

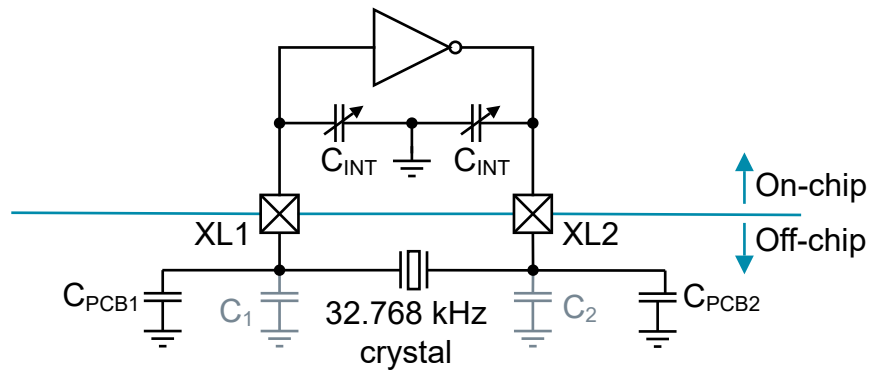


Figure 9: *LFXO*

The *LFXO* can be configured to use either the external capacitors C_1 and C_2 or the internal capacitors C_{INT} .

For more information, see [Low-frequency \(32.768 kHz\) crystal oscillator \(LFXO\)](#) in the Datasheet.

3.1 LFXO configuration with internal capacitors

When using internal capacitors, the load capacitance (C_L) is the total capacitance seen by the crystal across its terminals.

The following figure shows the 32.768 kHz crystal connected to the *LFXO* with internal capacitors.

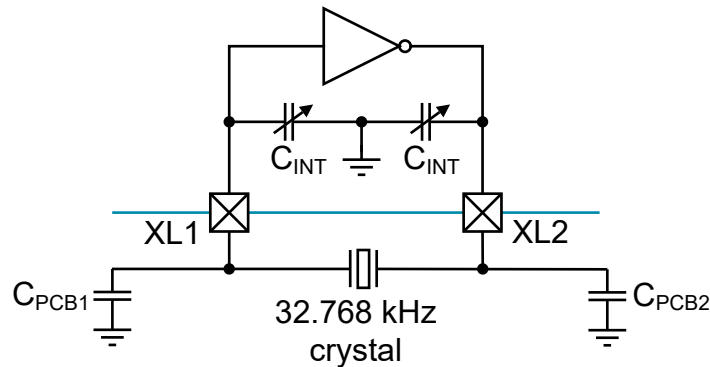


Figure 10: *LFXO* circuit diagram with internal capacitors only

The load capacitance for the internal capacitors is calculated with the following equation.

$$C_L = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C_{PIN} + C_{PCB1}$$

$$C2' = C_{PIN} + C_{PCB2}$$

Figure 11: Load capacitance equation for internal capacitors

C_{PCB1} and C_{PCB2} are stray capacitance on the PCB. C_{PIN} is the pin input capacitance on pins **XL1** and **XL2**, including C_{INT} and other capacitive loadings from the internal pad, on-chip ESD structure, and metal routes.

3.2 LFXO configuration with external capacitors

When using external capacitors, the load capacitance (C_L) is the total capacitance seen by the crystal across its terminals.

The load capacitance for the external capacitors is calculated with the following equation.

$$C_L = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C_1 + C_{PCB1} + C_{PIN}$$

$$C2' = C_2 + C_{PCB2} + C_{PIN}$$

Figure 12: Load capacitance equation for external capacitors

C_1 and C_2 are ceramic SMD capacitors connected between each crystal terminal and ground. C_{PCB1} and C_{PCB2} are stray capacitance on the PCB. C_{PIN} is the pin input capacitance on pins **XL1** and **XL2**.

It can be assumed that the capacitors are equal on each pin of the crystal. The external capacitor equation can then be simplified as follows:

$$C_1 = C_2 = 2 \cdot C_L - C_{stray}$$

C_L is the crystal's specified load capacitance. C_{stray} is the estimated stray capacitance from the PCB and package. C_{stray} is often in the range of 3 pF to 5 pF.

3.3 Select an LFXO crystal

Select an *LFXO* crystal that meets the specifications.

A simple model of a crystal is shown in [Select an HFXO crystal](#) on page 6.

The selected crystal must meet the required equivalent series resistance (ESR), motional inductance (L_M), and motional capacitance (C_M) specifications to ensure stable operation of the crystal oscillator.

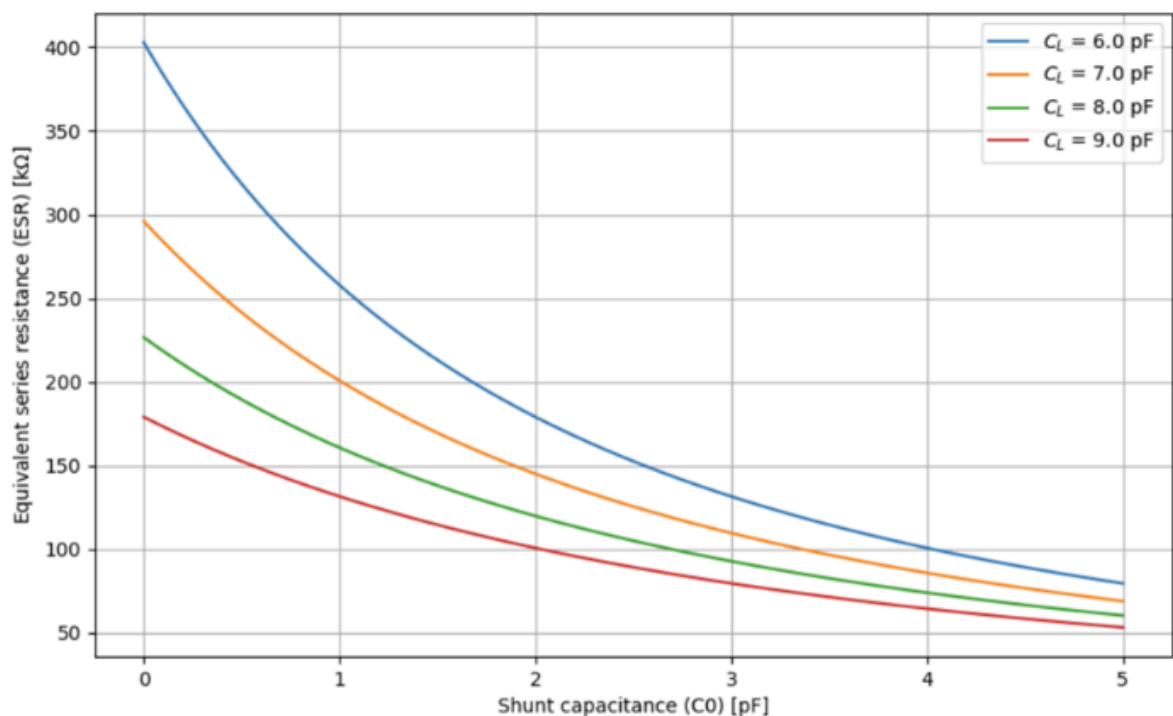
The external crystal should meet the specifications given in the datasheet of your device. The following is an example from nRF54L15 version 1.0.

Symbol	Description	Min.	Typ.	Max.	Units
f_{LFXO}	External crystal frequency.		32.768		kHz
$f_{\text{TOL_LFXO_BLE}}$	Frequency tolerance requirement for Bluetooth Low Energy applications.			± 500	ppm
$f_{\text{TOL_LFXO_ANT}}$	Frequency tolerance requirement for ANT [™] applications.			± 50	ppm
$C_{\text{L_LFXO}}$	Load capacitance.	6		9	pF
$C_{\text{O_LFXO}}$	Shunt capacitance.		1.0	2.0	pF
$R_{\text{S_LFXO}}$	Equivalent series resistance (ESR).		60	100	k Ω
$P_{\text{D_LFXO}}$	Drive level.			0.5	μW
C_{PIN}	Input capacitance on XL1 and XL2 pads with internal capacitor disabled (INTCAP=0).		3		pF

Table 5: External crystal specifications

If the crystal specifications do not match [Table 5: External crystal specifications](#) on page 16, check [Figure 13: Maximum allowed combinations of ESR and \$C_0\$ different load capacitance \$C_L\$ values](#) on page 16.

The following figure shows the maximum allowed combinations of ESR and C_0 for different load capacitance (C_L) values. A crystal is supported when its parameters fall on or below the load capacitance line of interest. A crystal is not supported when its parameters are above the line.

Figure 13: Maximum allowed combinations of ESR and C_0 different load capacitance C_L values

3.4 Startup time for selected LFXO crystal

To reduce startup time, use a low equivalent series resistance (ESR) and load capacitance (C_L).

The following table shows a typical simulated startup time for an external LFXO with an ESR of 60 k Ω , a load capacitance of 7 pF, and a shunt capacitance (C_0) of 1 pF.

Symbol	Description	Min.	Typ.	Max.	Units
$f_{\text{NOM_LFCLK}}$	Nominal output frequency		32.768		kHz
$t_{\text{START_LFXO}}$	Startup time for 32.768 kHz crystal oscillator		0.43		s

Table 6: External crystal specifications

3.5 Example case for LFXO crystal selection

The following example checks the parameters for a sample crystal to confirm that the nRF54L Series supports it. The example also shows how to calculate the load capacitance.

To check the crystal parameters, compare them with the general requirements given in [Table 5: External crystal specifications](#) on page 16. If the crystal does not match the specifications in the table, it needs to be further verified. For a sample crystal with a specified ESR, load capacitance (C_L), and maximum shunt capacitance (C_0), you can use the graph in [Figure 13: Maximum allowed combinations of ESR and \$C_0\$ different load capacitance \$C_L\$ values](#) on page 16 to determine if the crystal is supported. The crystal is supported if the parameter combination falls directly on the lines or below them.

For example, if the load capacitance in the crystal datasheet is 7.0 pF and ESR is 110 k Ω , the graph shows a maximum allowable shunt capacitance of about 3 pF. If the maximum shunt capacitance of the sample crystal is less than 3 pF, the crystal is suitable for the nRF54L Series.

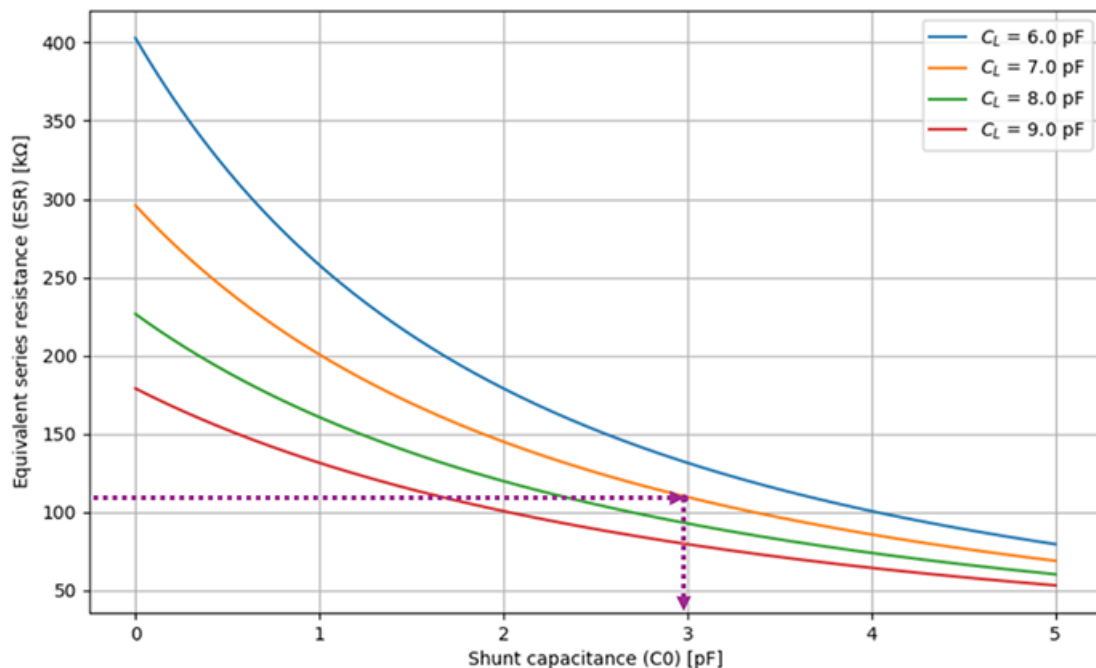


Figure 14: Sample crystal with 110 k Ω ESR and 7 pF C_L

Calculate the load capacitance for an [LFXO configuration with internal capacitors](#) on page 14 or [LFXO configuration with external capacitors](#) on page 15.

Assume equal capacitance on each pin and symmetrical PCB routing. Then, calculate the required load capacitors with the following formula.

$$C1 = C2 = 2 \cdot C_L - C_{\text{stray}}$$

C1 and C2 are the capacitors on the crystal pins. C_L is the load capacitance from the crystal datasheet. Cstray is the total capacitance on each crystal pin, including PCB trace capacitance and SoC pin capacitance.

For simplicity, use a value between 3–5 pF for Cstray. Use the higher end of the range when there are long crystal traces on the PCB.

After you calculate the required capacitance for each pin, decide whether to use either external or internal capacitors.

If you use external capacitors, place capacitors equal to the calculated values on each pin.

3.6 LFXO software configuration

The following sections describe how to correctly configure the software for the Low Frequency Crystal Oscillator (LFXO) on nRF54L Series devices. It focuses on ensuring correct load capacitor configuration.

The LFXO provides a precise 32.768 kHz clock source used for low-power operation, real-time clocks, and protocol timing. The LFXO is more accurate and stable than a fully internal RC oscillator (LFRC). However, it requires an external crystal, which is not needed for an LFRC.

3.6.1 LFXO devicetree

For a custom board, configure the LFXO settings in the devicetree using the &lfxo node.

The following example sets the load capacitor type and the internal capacitor value for a LFXO configuration.

```
&lfxo {
    load-capacitors = "internal";
    load-capacitance-femtofarad = <12000>;
};
```

The following table shows the configuration parameters.

Parameter	Function	Comment
load-capacitors	Set the LFXO load capacitor type to <code>internal</code> or <code>external</code> .	Set the load capacitor type to match your hardware design. It should match the actual crystal network on the PCB.
load-capacitance-femtofarad	Use to program the internal capacitor value when internal capacitors are used.	Use only when load-capacitors = "internal" . Determine the value based on crystal load capacitance and PCB stray capacitance.

Table 7: Devicetree configuration parameters

The **load-capacitors** parameter tells the firmware if the board uses the SoC's internal LFXO capacitors or external capacitors. Use the **load-capacitance-femtofarad** parameter only when internal capacitors are selected. This value should match the C_{INT} value chosen for the hardware design. External capacitors are fixed in hardware, so the software disables only the internal capacitors.

When you configure the LFXO software settings, ensure that the following guidelines are implemented:

- Update the internal load capacitor value. Do not copy the value from a *DK*.
- Consider the PCB stray and SoC pin capacitance when setting the value for the load capacitors.

If you change the crystal, capacitor mode, capacitor value, layout, voltage range, or temperature range on the board, review the software settings and update when needed.

3.7 Typical LFXO crystal specifications

The following table shows the crystal specifications used in the reference design for the nRF54L15 *DK*.

Item	Symbol	Value	Conditions
Nominal frequency	f_0	32.768 kHz	
Frequency tolerance	$\Delta f/f_0$	± 20 ppm	At 25°C
Load capacitance	C_L	9 pF	
Operating temperature	T_A	-40°C to 80°C	
Storage temperature range	T_{STR}	-55°C to 125°C	
Turnover temperature	T_M	25°C $\pm$ 5°C	
Temperature coefficient	β	-0.04 ppm/°C ² (max)	
Motional capacitance	C_M	6.5 pF (typ)	
Equivalent series resistance	ESR	90 k Ω (max)	At 25°C
Drive level	D_L	0.1 μ W (typ)	
Aging	$\Delta f/f_0$	± 3 ppm (max)	At 25°C
Shunt capacitance	C_0	1.3 pF (typ)	

Table 8: Electrical characteristics of crystal used in reference design for nRF54L15 *DK* (9HT11-32.768KDZC-T)

4 PCB layout design

Use the nRF54L Series reference designs as a basis for component values, geometry, relative placement of components, PCB stack-up, and trace lengths.

Documentation for the package reference circuits, including Altium Designer files, PCB layout files, and PCB production files can be downloaded from the product page on www.nordicsemi.com. See also the following resources:

- [nRF54L15 Reference layout](#)
- [nRF54L10 Reference layout](#)
- [nRF54L05 Reference layout](#)
- [nRF54LM20A Reference layout](#)
- [nRF54LV10A Reference layout](#)

4.1 Component placement

The reference designs in the datasheets are used for full validation of PCBs with the same external device placement and metal routings. Use the same or a similar approach to create your own design.

4.2 PCB stack-up

Follow the reference design as close as possible for the PCB stack-up.

For the nRF54L Series, the following targets should be considered:

- Use PCB stack-up based on the nRF54L Series reference designs.
- Use four or more metal layers to achieve the full radio output power dynamic range.
- Keep prepreg thickness, which is the dielectric between the top metal and the first mid-layer, at less than 100 μm to reduce parasitic inductance of the via between the top and mid-layers.
- Use thinner prepreg when possible to reduce effective inductance and avoid negative impact on radio performance.

The following figure shows the prepreg and core for a typical four-layer PCB stack-up.

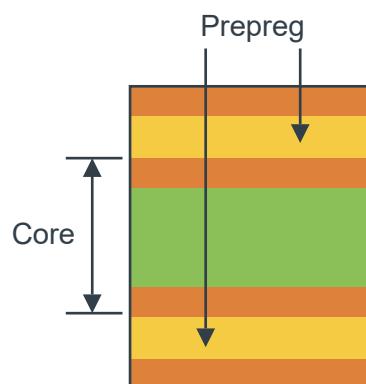


Figure 15: Typical four-layer PCB stack-up

4.3 DECRF routing

Special care should be taken when routing the RF supply input (DECRF), the DC/DC regulator output, and the **XC1** and **XC2** pins.

The crystal oscillator provides a clean reference clock used throughout nRF54L Series devices for both radio and digital functions. In the nRF54L Series, the 32 MHz crystal pins **XC1** and **XC2** are positioned between the DC/DC converter output and the RF supply input pin **DECRF**. **XC1** and **XC2** must be connected by a PCB trace. When routing these connections, follow the reference layout and keep the tracks as short as possible to prevent crosstalk between the crystal and the RF supply.

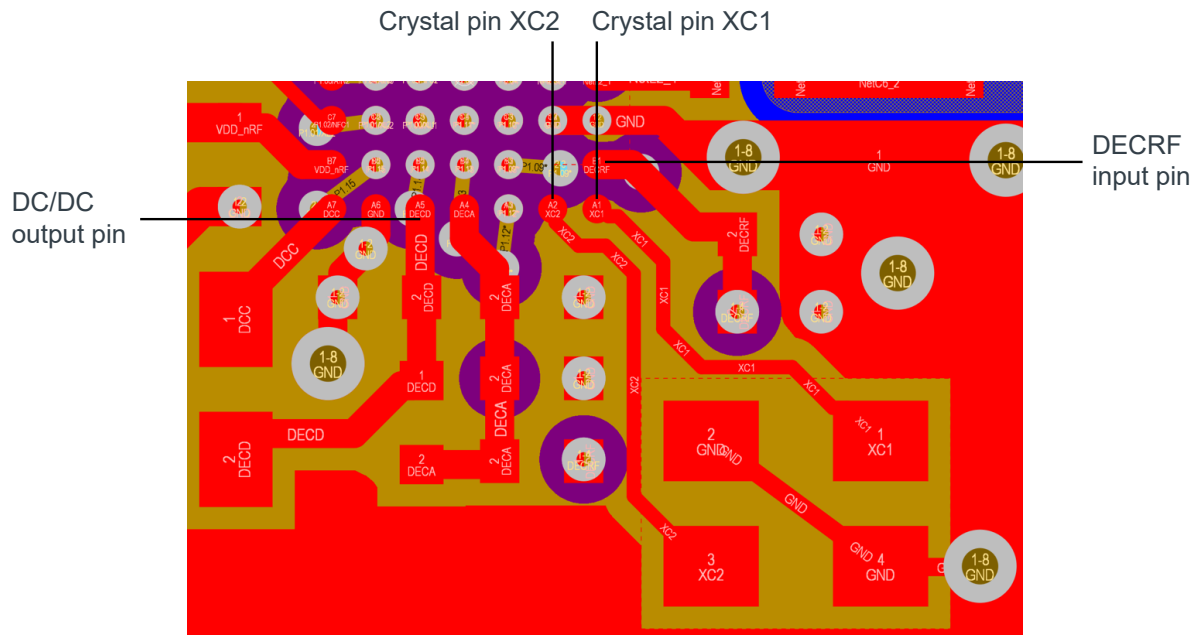


Figure 16: Crystal pins XC1 and XC2 location between the DC/DC output pin and DECRF input pin on the CSP package

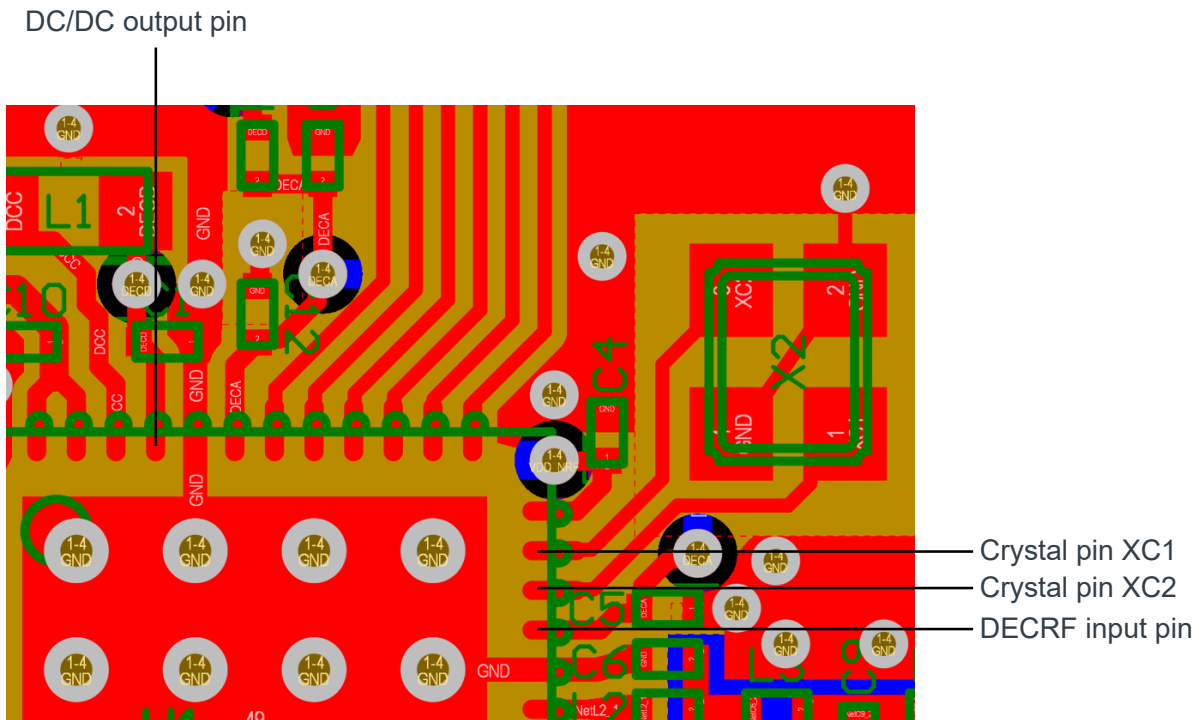


Figure 17: Crystal pins XC1 and XC2 location between the DC/DC output pin and DECRF input pin on the QFN package

4.4 Shunt capacitor for QFN

The RF matching network uses a three-stage L-C network.

In the QFN reference design, the second shunt capacitor is placed on its own ground island and connects to the bottom ground layer through three PCB vias. This island also connects to the die center pad through an array of vias down to the bottom plane. Together, this structure creates a high-frequency series resonance that helps suppress transmitter harmonics. The exact notch frequency can shift depending on the PCB stack-up.



Follow the reference design as close as possible when placing the crystal and its metal connections.

- ## 4.6 LFXO crystal placement and routing

- Keep traces between the nRF54L Series SoC and the crystal as short as possible to reduce parasitic capacitance and noise.
- Route **XL1** and **XL2** symmetrically and avoid placing them near high-speed or noisy signals.
- Place a solid ground plane under the crystal and its traces to shield from interference.
- Consider that the layout can significantly affect the total capacitance seen by the crystal.

4503 586

Glossary

Development Kit (DK)

A hardware development platform used for application development.

HFXO

High-Frequency Crystal Oscillator

LFXO

Low-Frequency Crystal Oscillator

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

Recommended reading

In addition to the information in this document, you may need to consult other documents.

Nordic documentation

- [nRF54L15 | nRF54L10 | nRF54L05 Datasheet](#)
- [nRF54LM20A | nRF54LM20B Datasheet](#)
- [nRF54LS05A | nRF54LS05B Datasheet](#)
- [nRF54LV10A Preliminary Datasheet](#)

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